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# Online Examination Report

|                |                                |
|----------------|--------------------------------|
| Date:          | 21.07.2014                     |
| Attempts:      | 1                              |
| Examinee:      | Anoku, Martin                  |
| Date of birth: | 8/28/1984                      |
| Subject:       | 030-Flight Performance&Planing |
| Result:        | 38,1 %                         |
| Time used:     | 00:00:12 (HH:MM:SS)            |

## Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

## Attachment 1: List of result by question

| CQB Number | Internal Number | Syllabus reference | Chapter                                   | Points | Max. Points |
|------------|-----------------|--------------------|-------------------------------------------|--------|-------------|
| 1          | 388             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 2          | 374             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 3          | 378             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 4          | 373             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 5          | 390             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 6          | 375             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 1,00   | 1,00        |
| 7          | 391             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 8          | 382             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 9          | 381             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 10         | 361             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 11         | 315             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 12         | 379             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 1,00   | 1,00        |
| 13         | 380             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 1,00   | 1,00        |
| 14         | 183             | 31.2               | LOADING                                   | 0,00   | 1,00        |
| 15         | 278             | 31.3.3             | Basic calculations of CG                  | 0,00   | 2,00        |
| 16         | 424             | 32.                | PERFORMANCE - AEROPLANES                  | 0,00   | 1,00        |
| 17         | 339             | 32.                | PERFORMANCE - AEROPLANES                  | 1,00   | 1,00        |
| 18         | 428             | 32.                | PERFORMANCE - AEROPLANES                  | 1,00   | 1,00        |
| 19         | 20              | 32.1.2.2           | Definitions, Terms and Concepts           | 1,00   | 1,00        |
| 20         | 106             | 32.1.2.2           | Definitions, Terms and Concepts           | 0,00   | 1,00        |
| 21         | 77              | 32.2.5.4           | Landing                                   | 1,00   | 1,00        |
| 22         | 231             | 32.2.5.4           | Landing                                   | 0,00   | 1,00        |
| 23         | 115             | 32.4.1             | Take - off                                | 0,00   | 1,00        |
| 24         | 146             | 32.4.1.7           | Take-off climb                            | 0,00   | 1,00        |
| 25         | 219             | 32.4.1.7           | Take-off climb                            | 1,00   | 1,00        |
| 26         | 135             | 32.4.1.7           | Take-off climb                            | 1,00   | 1,00        |
| 27         | 189             | 32.4.1.7           | Take-off climb                            | 1,00   | 1,00        |
| 28         | 167             | 32.4.3             | Cruise                                    | 1,00   | 1,00        |

| CQB Number       | Internal Number | Syllabus reference | Chapter                                                                | Points       | Max. Points  |
|------------------|-----------------|--------------------|------------------------------------------------------------------------|--------------|--------------|
| 29               | 202             | 32.4.3             | Cruise                                                                 | 1,00         | 1,00         |
| 30               | 82              | 32.4.3             | Cruise                                                                 | 1,00         | 1,00         |
| 31               | 92              | 33.                | FLIGHT PLANNING AND FLIGHT MONITORING                                  | 0,00         | 1,00         |
| 32               | 417             | 33.1               | FLIGHT PLANNING FOR VFR FLIGHTS                                        | 0,00         | 1,00         |
| 33               | 148             | 33.1.1.1           | Routes, airfields, heights and altitudes from VFR charts               | 0,00         | 1,00         |
| 34               | 221             | 33.1.1.1           | Routes, airfields, heights and altitudes from VFR charts               | 1,00         | 1,00         |
| 35               | 259             | 33.1.1.5           | Completion of navigation plan                                          | 0,00         | 2,00         |
| 36               | 122             | 33.3.3             | Specific fuel calculation procedures                                   | 0,00         | 1,00         |
| 37               | 252             | 33.4.1.3           | Airway routings and airspace structure                                 | 1,00         | 1,00         |
| 38               | 224             | 33.4.2.1           | Extraction and analysis of relevant data from meteorological documents | 0,00         | 1,00         |
| 39               | 203             | 33.5.1.1           | Format of Flight Plan                                                  | 1,00         | 1,00         |
| 40               | 30              | 33.5.1.1           | Format of Flight Plan                                                  | 0,00         | 1,00         |
| <b>Total:38%</b> |                 |                    |                                                                        | <b>16,00</b> | <b>42,00</b> |

## Attachment 2: List of result by topic

**Licence**

**Date: 18.07.2014**

**Examinee: Anoku, Martin**

**Location: Uganda Civil Aviation Authority**

**Your Result: 16,00 from 42,00 Points (38,10 %)**

| Topics                                                               | Ammount of Questions | Points achieved | Maximum Points | Percent      |
|----------------------------------------------------------------------|----------------------|-----------------|----------------|--------------|
| 31. MASS AND BALANCE - AEROPLANES/HELICOPTERS                        | 15                   | 3,00            | 16,00          | 18,75        |
| 31.2. LOADING                                                        | 1                    | 0,00            | 1,00           | 0,00         |
| 31.3. FUNDAMENTALS OF CG CALCULATIONS                                | 1                    | 0,00            | 2,00           | 0,00         |
| 32. PERFORMANCE - AEROPLANES                                         | 15                   | 10,00           | 15,00          | 66,67        |
| 32.1. GENERAL                                                        | 2                    | 1,00            | 2,00           | 50,00        |
| 32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES                 | 2                    | 1,00            | 2,00           | 50,00        |
| 32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY | 8                    | 6,00            | 8,00           | 75,00        |
| 33. FLIGHT PLANNING AND FLIGHT MONITORING                            | 10                   | 3,00            | 11,00          | 27,27        |
| 33.1. FLIGHT PLANNING FOR VFR FLIGHTS                                | 4                    | 1,00            | 5,00           | 20,00        |
| 33.3. FUEL PLANNING                                                  | 1                    | 0,00            | 1,00           | 0,00         |
| 33.4. PRE-FLIGHT PREPARATION                                         | 2                    | 1,00            | 2,00           | 50,00        |
| 33.5. ICAO FLIGHT PLAN (ATS Flight Plan)                             | 2                    | 1,00            | 2,00           | 50,00        |
| <b>Total</b>                                                         | <b>40</b>            | <b>16,00</b>    | <b>42,00</b>   | <b>38,10</b> |

1

**What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?**

- ☐ Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.
- ☐ Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.
- ☒ There is no relationship between CG location and indicated airspeed.

Candidate: Martin Anoku  
Date of birth: 1984-08-28

Subject: 030-Flight Performance&Planing  
Date: 2014-07-18



2

**How much weight could be added at Station 120 without exceeding the aft CG limit?**

**Aircraft weight 9,500 lb**

**CG location Station 90.0**

**Aft CG limit Station 90.5**

☐ 61.0 pounds.

☐ 161.0 pounds.

☒ 110.5 pounds.

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3

**What is the location of the CG if 60 pounds are removed from Station 70?**

**Aircraft weight 8,420 lb**

**CG location Station 85**

☐ 84.9.

☒ 84.1.

☐ 85.1.

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4

**How much weight could be added at Station 160 without exceeding the aft CG limit?**

**Aircraft weight 8,300 lb**

**CG location Station 90.0**

**Aft CG limit Station 90.5**

☒ **16.5 pounds.**

☐ **13.9 pounds.**

☐ **59.7 pounds.**

5

**Under which condition is a forward CG most critical?**

☐ When in an unusual attitude.

☐ On landing.

☒ On takeoff.

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6

What is the maximum weight that could be added at Station 150.0 without exceeding the aft CG limit?

Aircraft weight 5,000 lb  
CG location Station 80.0

☒ 35.9 pounds.

☐ 69.5 pounds.

☐ 70.0 pounds.

7

As the CG location is changed, recovery from a stall becomes progressively

- ☒ less difficult as the CG moves rearward.
- ☐ more difficult as the CG moves either forward or rearward.
- ☐ more difficult as the CG moves rearward.

8

The center of gravity of an aircraft can be determined by

- ☐ dividing total moment by total weight.
- ☐ dividing total arm by total moments.
- ☒ multiplying total arm by total weight.

9

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☐ Yes; the CG would remain at Station 115.8.

☐ Yes; the new CG would be located at Station 117.69.

☒ No; the new CG would be located at Station 118.15.

10

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?  
Floor load limit - 85 lb/sq ft  
Pallet weight - 44 lb  
Tiedown devices - 29 lb

☐ 924.5 pounds.

☒ 953.6 pounds.

☐ 968.6 pounds.

11

The term 'weigh-off' as used in ballooning means to determine the

- ☐ amount of fuel required for ascent to a preselected altitude.
- ☐ static equilibrium of the balloon as loaded for flight.
- ☒ standard weight and balance of the balloon.

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12

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☐ 68.9 pounds.

☒ 62.5 pounds.

☐ 65.8 pounds.

13

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight      2,800 lb  
CG location      Station 120.0

☐ Yes; the new CG would be located at Station 117.89.

☒ No; the new CG would be located at Station 116.42.

☐ No; the new CG would be located at Station 116.89.

14

*In flight the wing of an aircraft containing fuel is subjected to vertical loads that produce a bending moment which is:*

- ☐ highest at the wing root
- ☐ equal to half the weight of the aircraft multiplied by the semi span
- ☐ equal to the zero -fuel weight multiplied by the span
- ☒ lowest at the wing root

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15

(Please view annex 031-038 rev. 05.11.2004)  
For the following boom to be in balance:

☐  $A = B \times Fb / Fa$

☐  $A = B \times Fa / Fb$

☐  $A = B (Fa + Fb)$

☒  $A = B + Fb / Fa$

See Attachments:

1. 031-038.gif

Annex

(Attachment  
No.1)

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16

To level off from a descent maintaining the descending airspeed, the pilot should lead the desired altitude by approximately

☒ 20 feet.

☐ 60 feet.

☐ 50 feet.

17

**(Refer to Figure 89.) Which of the following configurations will result in the shortest landing distance over a 50-foot obstacle to a wet runway?**

See Attachments:

1. atp\_89.jpg

figure 89

(Attachment  
No.2)

- ☐ Brakes and spoilers at 122,500 pounds gross weight.
- ☒ Brakes, spoilers, and reversers at 131,000 pounds gross weight.
- ☐ Brakes and reversers at 124,000 pounds gross weight.

18

If while in level flight, it becomes necessary to use an alternate source of static pressure vented inside the airplane, which of the following variations in instrument indications should the pilot expect?

- ☐ The vertical speed to show a climb.
- ☒ The altimeter to read higher than normal.
- ☐ The vertical speed to momentarily show a descent.

19

Besides lift, the forces that determine the gradient of climb of an aeroplane are:

- ☐ *Weight and thrust only.*
- ☒ *Weight, drag and thrust.*
- ☐ *Thrust and drag only.*
- ☐ *Weight and drag only.*

20

Consider the graphic representation of the power required versus true air speed (TAS), for a jet aeroplane with a given mass. When drawing the tangent out of the origin, the point of contact determines the speed of:

- ☐ increased
- ☐ decreased
- ☐ unchanged
- ☒ very significantly decreased

21

A runway is contaminated with 0.5 cm of wet snow. The flight manual of a light twin nevertheless authorises a landing in these conditions. The landing distance will be, in relation to that for a dry runway:

☐ unchanged

☐ reduced

☒ increased

☐ substantially decreased

22

To minimize the risk of hydroplaning during landing the pilot should:

- ☐ use maximum reverse thrust, and should start braking below the hydroplaning speed.
- ☐ postpone the landing until the risk of hydroplaning no longer exists.
- ☐ make a "positive" landing and apply maximum reverse thrust and brakes as quickly as possible.
- ☒ use normal landing-, braking- and reverse technique.

23

**When the outside air temperature increases, then**

- ☐ the field length limited take-off mass decreases but the climb limited take-off mass increases.
- ☐ the field length limited take-off mass and the climb limited take-off mass decreases.
- ☐ the field length limited take-off mass and the climb limited take-off mass increases.
- ☒ the field length limited take-off mass increases but the climb limited take-off mass decreases.

24

**If the take-off mass of an aeroplane is brake energy limited a higher uphill slope would**

- ☐ decrease the required take-off distance.
- ☒ decrease the maximum mass for take-off.
- ☐ have no effect on the maximum mass for take-off.
- ☐ increase the maximum mass for take-off.

25

**During climb to the cruising level, a headwind component**

- ☒ decreases the ground distance flown during that climb.
- ☐ increases the climb time.
- ☐ increases the amount of fuel for the climb.
- ☐ decreases the climb time.

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26

With an true airspeed of 194 kt and a vertical speed of 1 000 ft/min, the climb gradient is about :

☐ 0.03

☐ 0.08

☒ 3°

☐ 5°

27

The climb limited take-off mass can be increased by

- ☒ a lower flap setting for take-off and selecting a higher V2.
- ☐ selecting a lower V2.
- ☐ selecting a lower VR.
- ☐ selecting a lower V1.

28

**Density altitude is the**

- ☐ height above the surface
- ☐ altitude reference to the standard datum plane
- ☒ pressure altitude corrected for 'non standard' temperature
- ☐ altitude read directly from the altimeter

29

For a piston engined aeroplane, the speed for maximum range is :

- ☐ that which gives the maximum value of lift
- ☐ that which gives the minimum value of drag.
- ☒ that which gives the maximum lift to drag ratio.
- ☐ 1.4 times the stall speed in clean configuration.

30

Which of the following statements is correct?

- ☐ Induced drag decreases with increasing angle of attack.
- ☒ Induced drag decreases with increasing speed.
- ☐ Induced drag is independant of the speed.
- ☐ Induced drag increases with increasing speed.

31

From which of the following would you expect to find details of the Search and Rescue organisation and procedures (SAR) ?

☐ SIGMET

☒ AIP (Aeronautical Information Publication)

☒ NOTAM

☐ ATCC broadcasts

32

**How long will a Flight Service Station hold a VFR flight plan past the proposed departure time?**

☐ 1 hour.

☐ 30 minutes.

☒ 2 hours.

33

From which of the following would you expect to find information regarding known short unserviceability of VOR, TACAN, and NDB ?

☐ D ATCC broadcasts

☒ A NOTAM

☐ C SIGMET

☒ B AIP (Aeronautical Information Publication)

34

**VFR flights shall not be flown over the congested areas of cities at a height less than**

- ☐ B 2000 ft above the heighest obstacle within a radius of 600 ft from the aircraft.
- ☐ D the heighest obstacle.
- ☒ A 1000 ft above the heighest obstacle within a radius of 600 m from the aircraft.
- ☐ C 500 ft above the heighest obstacle.

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35

Given: true course (TC) 017; W/V 340/30; TAS 420 kt  
Find: wind correction angle (WCA) and ground speed (GS)

☐ WCA +2° ; GS 396 kt

☒ WCA +2° ; GS 416 kt

☐ WCA -2° ; GS 426 kt

☐ WCA -2° ; GS 396 kt

36

An aircraft flies at an airspeed of 380 kt. It flies from A to B and back to A. Distance AB = 480 NM. When going from A to B, it experiences a headwind component = 60 kt. The wind remains constant. The duration of the flight will be:

☐ 2h 32min

☒ 3h 00min

☐ 2h 10min

☐ 2h 35min

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37

From which of the following would you expect to find the dates and times when temporary danger areas are active

☐ **SIGMET**

☐ **Only AIP**

☒ **NOTAM and AIP**

☐ **RAD/NAV charts**

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38

A METAR reads : SA1430 35002KT 7000 SKC 21/03 Q1024 =  
Which of the following information is contained in this METAR?

- ☐ *runway in use*
- ☐ *period of validity*
- ☒ *day/month*
- ☐ *temperature/dewpoint*

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39

When completing an ATC flight plan, an elapsed time (Item 16) of 1 hour 55 minutes should be entered as:

☒ 0155

☐ 115M

☐ 0115

☐ 1H55

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40

If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

☐ D ZZZZ

☐ A AAAA

☐ C IIII

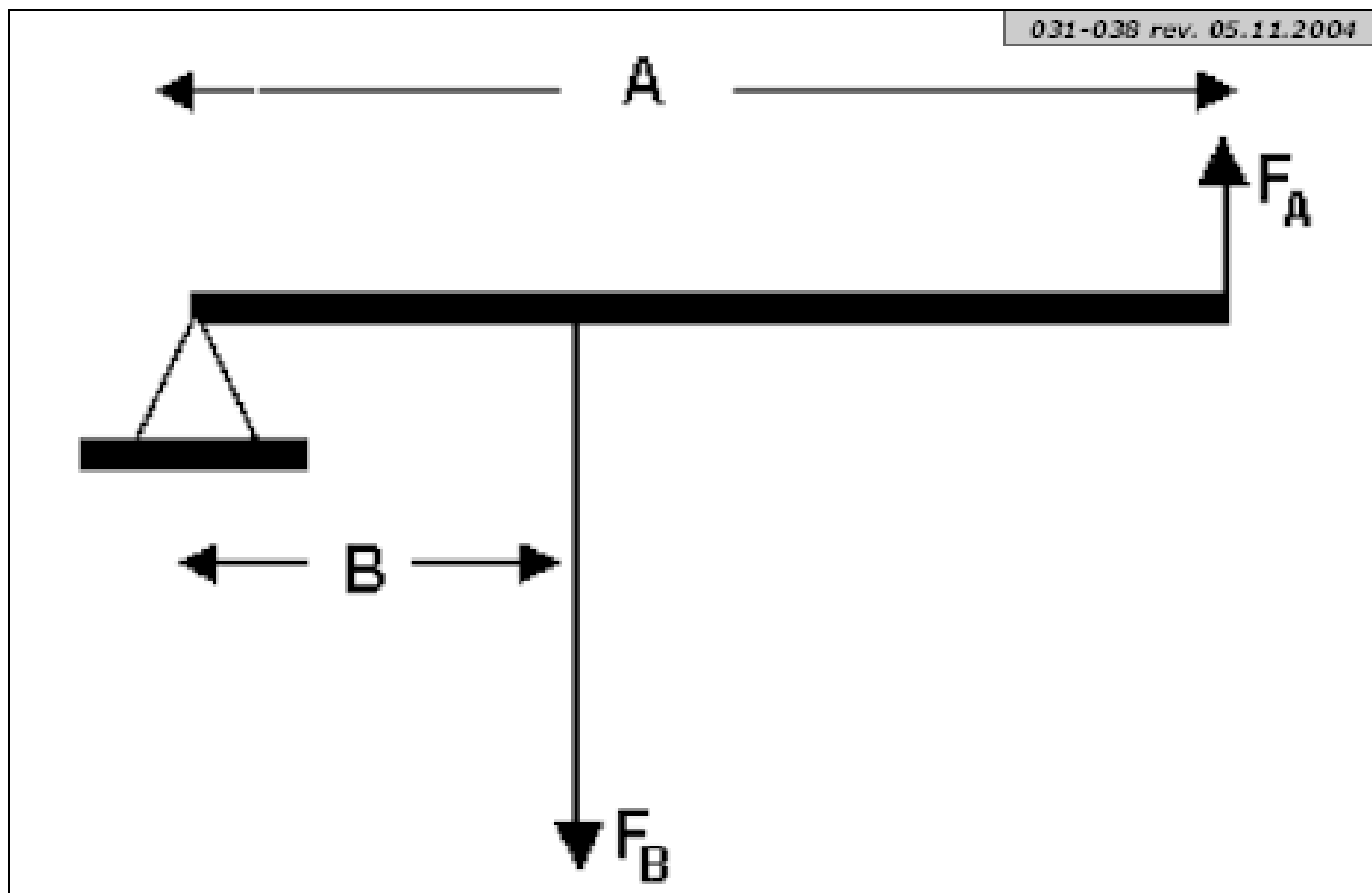
☒ B XXXX

## Attachment No. 1

Questions 15

Annex

031-038.gif

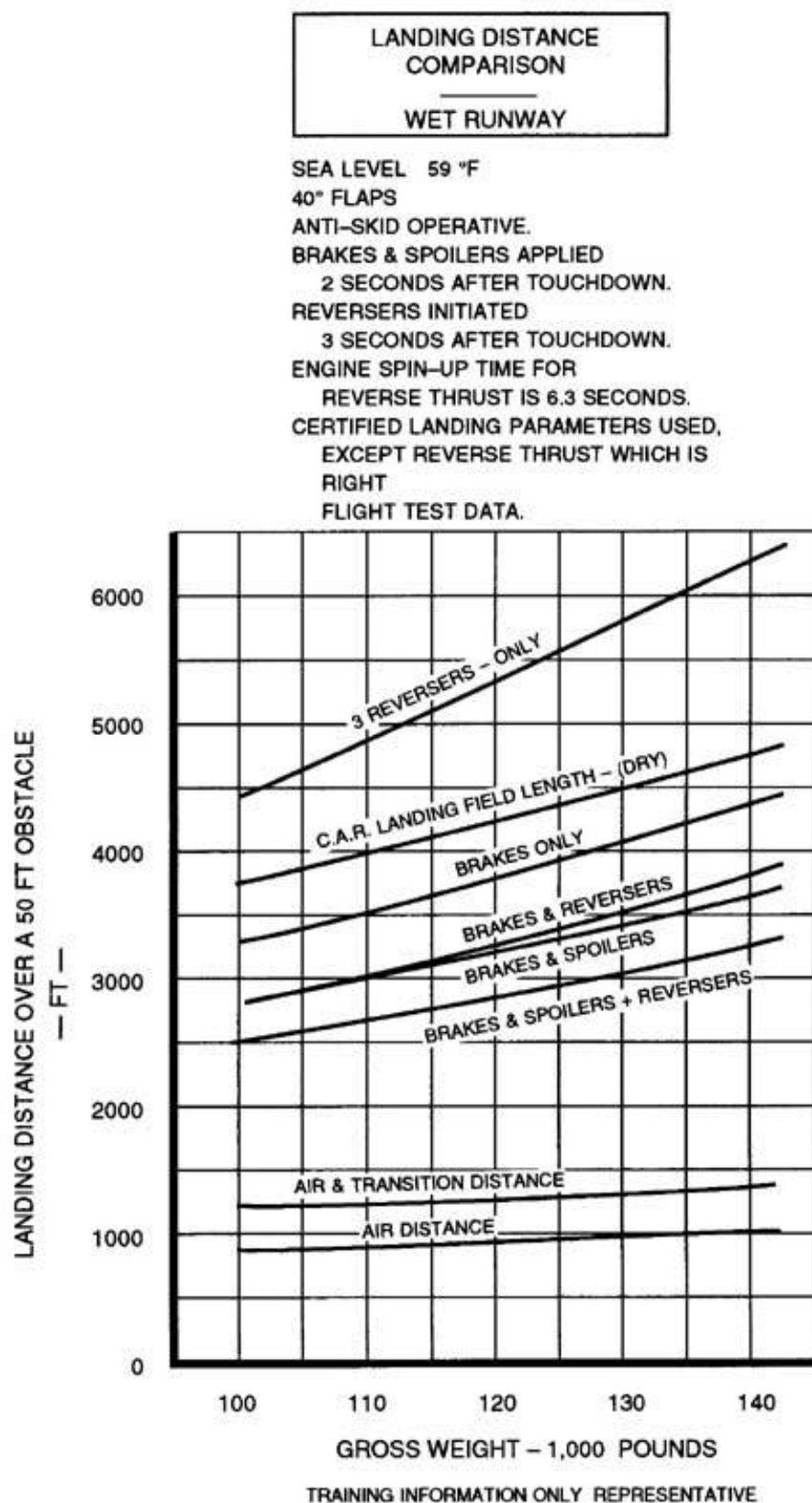


## Attachment No. 2

Questions 17

figure 89

atp\_89.jpg



### NORMAL LANDING

**Figure 89. B-727 – Normal Landing – Wet Runway © ASA**

